

INTERNATIONAL ISLAMIC UNIVERSITY CHITTAGONG

LAB MANUAL

Computer Science and Engineering Department

1. **Course Code:** CSE 2324
2. **Title of the course:** Digital Logic Design Lab
3. **Credit Hour:** 1.50 credits
4. **Contact Hours:** 3 Hour/Week
5. **Prerequisite:** N/A
6. **Co-requisite:** CSE-2323(Digital Logic Design)

7. **Rationale:**

In this lab course students will develop practical design skills and become familiar with basic digital hardware by constructing different combinational circuits, and programmable logic devices. Finally, they apply their knowledge to the design of practical circuits such as alarm system, stopwatch etc. Students also deal with different matters such as the course objectives, laboratory rules and procedures, safety issues, preparation of pre-lab and lab reports.

8. **Objectives:** This course is concerned with the practical aspects of designs based on the course CSE-2323, Digital logic design theory.

9. Course Outcomes (COs):

CO1	Understand the usage of the tools for implementing simple, combinational and sequential logic circuit.
CO2	Implement different efficient digital logic circuits for different logical problems.
CO3	Design small project as a team member to measure and record the experimental data, analyze the results, and prepare a formal laboratory report.

10. Mapping of CO-PO:

Cos	Corresponding POs	Bloom's taxonomy domain/level (C:Cognitive, P:Psychomotor A: Affective)	Delivery methods and activities	Assessment tools
CO1	PO1	Cognitive/Understand	Lecture & Laboratory Experiments	Quiz & Performance Test
CO2	PO4	Cognitive/Apply	Lecture & Laboratory Experiments	Quiz & Performance Test
CO3	PO3	Cognitive/Analyze	Lecture & Laboratory Experiments	Quiz & Performance Test

11. Resources:**Text Books:**

#	Name of Authors	Title of Book	Edition	Publisher's Name	Year	ISBN
1.	M. Morris Mano	Digital Logic and Computer Design	5th Edition	Pearson education	2013	ISBN-13: 978-0-13-277421-5

Reference Books:

#	Name of Authors	Title of Book	Edition	Publisher's Name	Year	ISBN
1.	Md. Mozammel Huq Azad Khan	Digital logic design	1 st Edition	University Grants Commission of Bangladesh	2006	ISBN-13: 978-9848090244
2.	Thomas L. Floyd	Digital Fundamentals	11 th Edition	Pearson	2014	ISBN-13: 978-0132737968
3.	Tocci-widmer	Digital Systems	12 th Edition	Pearson Prentice Hall	2016	ISBN-13: 978-0134220130

12. Weightage Distribution among Assessment Tools:

Assessment Tools	Weightage (%)
Class Attendance	10
Continuous Assessment	40
Final Examinations	50

13. Grading Policy:

As per IIUC grading policy

14. Weekly Activity Plan & List of Experiments:

Week	Activities	Experiment Name
1.	Lab Work	Investigating the Logic Gates: - Fundamental Logic Gate - AND, OR, NOT. - Fundamental Logic Gate - NAND and NOR.
2.	Lab Work	Deducing truth tables from logic functions: - Deduce the truth table for the given switching circuit. Find the Boolean function from the truth table and implement the logic circuit. - An alarm system will ring when smoke is detected or movement is detected during night only. Design and implement the logic circuit. If two numbers A and B are added, whether the summation is odd or even can always be determined from the LSB of the two numbers. Design and implement a circuit, which will take the LSB of two numbers as inputs and determine whether the addition of these two will produce an odd or even number.
3.	Lab Work	Using De Morgan: Design a circuit to verify De Morgan's Law for two variables. Design a circuit using only a) NAND gates b) NOR gates that implements the Boolean function $f(A, B, C) = (A \oplus B) + (BC)$
4.	Lab Work	Derive the output equations for a 3 Boolean Function Minimization: - Design and implement a circuit using basic gates that would take a 4-bit binary as input and produce a high level output only if the number is prime or divisible by five. - Design a circuit to implement the following functions (Minimize if necessary) $f(A, B, C, D) = \sum(2, 3, 4, 6, 7, 10, 11, 14)$ $f(A, B, C, D) = \prod(1, 3, 5, 7, 9, 11)$ 4-bit binary to gray code converter and implement it.
5.	Lab Work	Combinational Circuits: Design and implement a circuit using basic gates that would take a 4-bit binary number as input and convert it into Excess - 3 Design and implement a 4-bit adder subtractor circuit with one selection variable to switch for addition and subtraction.
6.		Mid Term Exam

7.	Lab Work	Adders and Decoders a. Design and implement a 4-bit BCD adder circuit using 4-bit binary adder circuit. Use IC-7483 for your design. Design a 2-bit comparator to compare 2-bit numbers A and B. The circuit should provide 3 output lines to indicate $A > B$, $A = B$ and $A < B$.
8.	Lab Work	Priority Encoder and Multiplexer Circuits: a. Using basic gates design and implement a 4×2 priority encoder with the priority of data bit input as given below: (2, 3, 0, 1 given in order of descending priority) b. Design and implement the function given below using 1) 8:1 Multiplexer and 2) 4:1 Multiplexer $f = \sum (0, 3, 6, 8, 12)$
9.	Lab Work	Sequential Circuit Design: a. Design and implement an S-R flip-flop using only NAND gates. There must be PRESET and CLEAR input. Design and implement a master-slave J-K flip-flop using only NAND gates.
10.	Lab Work	Counter Circuit Design: a. Design and implement a 4-bit asynchronous counter using J-K flip flop. Design and implement a modulo-13 counter using 4-bit binary ripple counter (7493).
11.	Lab Work	Stopwatch and Counter Circuit Design a. Design and implement a digital stopwatch showing only seconds. Use 7-segment display to show the stopwatch output. You should also provide START, STOP and CLEAR switch. Design and implement a 4-bit synchronous up-down counter using J-K flip-flops.
12.	Lab Work	Stopwatch and Counter Circuit Design Continued.
13.	Lab Work	Projects
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Experiment1:

Experiment Name: Investigating the Logic Gates:

a. Fundamental Logic Gate - AND, OR, NOT.

b. Fundamental Logic Gate - NAND and NOR.

Aim:-

Introduction to digital electronics lab- nomenclature of digital ICs, specifications, study of the data sheet, concept of Vcc and ground, verification of the truth tables of logic gates using TTL ICs.

Apparatus Required:-

Digital lab kit, single strand wires, breadboard, TTL IC's

Gates	IC NO.
AND	7408
OR	7432
NAND	7400
NOR	7402
NOT	7404

Theory:-

Logic gates are idealized or physical devices implementing a Boolean function, which it performs a logical operation on one or more logical inputs and produce a single output. Depending on the context, the term may refer to an ideal logic gate, one that has for instance zero rise time and unlimited fan out or it may refer to a non-ideal physical device.

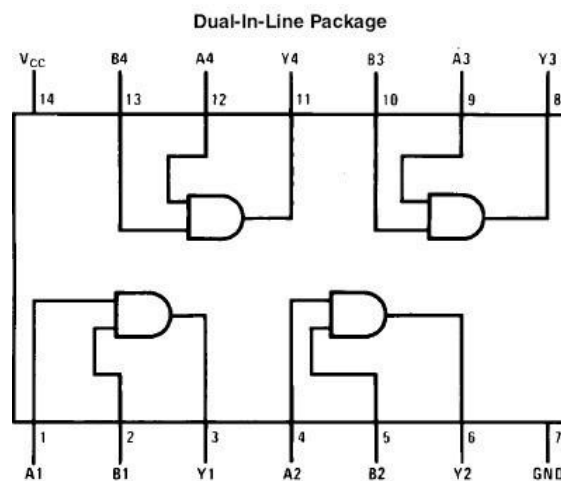
The main hierarchy is as follows:-

1. Basic Gates
2. Universal Gates

Basic Gates:

1. **AND gate:** - Function of AND gate is to give the output true when both the inputs are true. In all the other remaining cases output becomes false. Following table justifies the statement:-

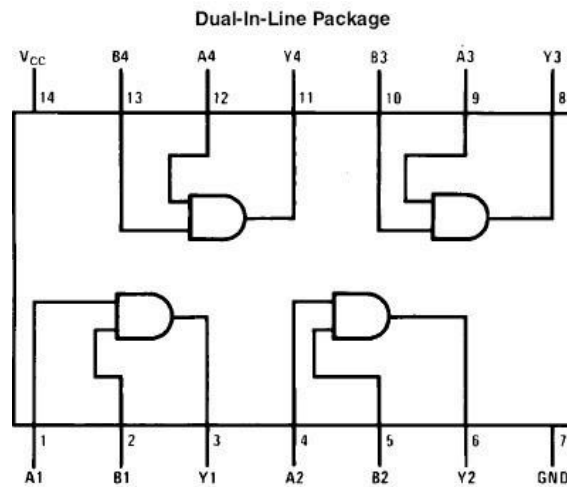
Input A	Input B	Output
1	1	1
1	0	0
0	1	0
0	0	0



IC 7408

2. **OR gate:** - Function of OR gate is to give output true when one of the either inputs are true .In the remaining case output becomes false. Following table justify the statement:-

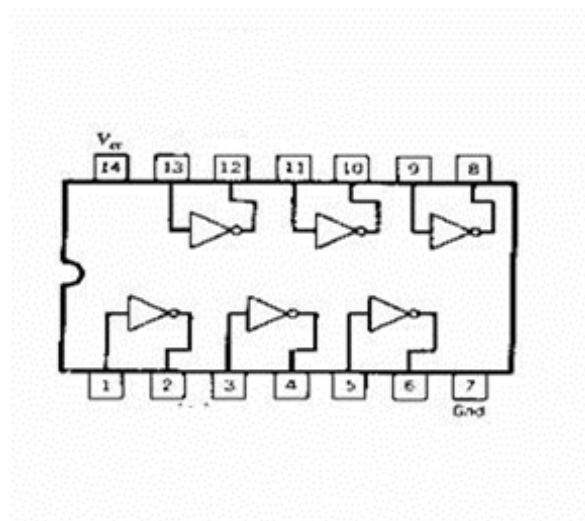
Input A	Input B	Output
0	0	0
0	1	1
1	0	1
1	1	1



IC 7432

3. **NOT gate:** - Function of NOT gate is to reverse the nature of the input. It converts true input to false and vice versa. Following table justifies the statement :-

4. Input	Output
1	0
0	1



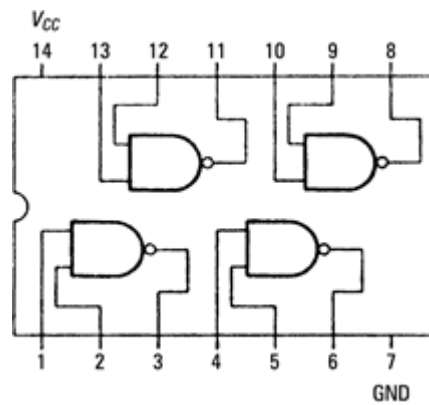
IC 7404

Universal Gates:

1. **NAND gate:** - Function of NAND gate is to give true output when one of the two provided input are false. In the remaining output is true case. Following table justifies the statement :-

Input A	Input B	Output
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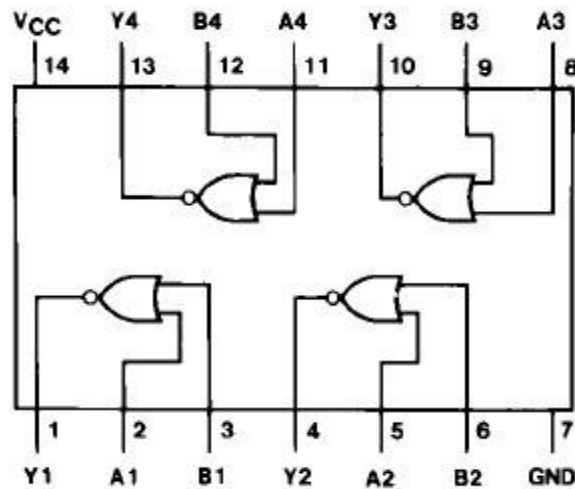
1	1	0
1	0	1
0	1	1
0	0	1



IC 7400

2. **NOR gate:** - NOR gate gives the output true when both the two provided input are false. In all the other cases output remains false. Following table justifies the statement:-

Input A	Input B	Output
1	1	0
1	0	0
0	1	0
0	0	1



IC 7402

Procedure:-

- Place the breadboard gently on the observation table.
- Fix the IC which is under observation between the half shadow line of breadboard, so there is no shortage of voltage.
- Connect the wire to the main voltage source (V_{CC}) whose other end is connected to last pin of the IC (14 place from the notch).
- Connect the ground of IC (7th place from the notch) to the ground terminal provided on the digital lab kit.
- Give the input at any one of the gate of the ICs i.e. 1st, 2nd, 3rd, 4th gate by using connecting wires.(In accordance to IC provided).
- Connect output pins to the led on digital lab kit.
- Switch on the power supply.
- If led glows red then output is true, if it glows green output is false, which is numerically denoted as 1 and 0 respectively. The Color can change based on the IC manufacturer it's just verification of the Truth Table not the color change.

Result:-

Experiment No: 02

Experiment name:

Deducing truth tables from logic functions:

Deduce the truth table for the given switching circuit. Find the Boolean function from the truth table and implement the logic circuit.

An alarm system will ring when smoke is detected or movement is detected during night only. Design and implement the logic circuit.

If two numbers A and B are added, whether the summation is odd or even can always be determined

from the LSB of the two numbers. Design and implement a circuit, which will take the LSB of two numbers as inputs and determine whether the addition of these two will produce an odd or even number.

AIM:

To deduce the truth table for the given switching circuit. Find the Boolean function from the truth table and implement the logic circuit.

Apparatus Required:-

- IC 7400
- IC 7408
- IC 7432
- IC 7406
- IC 7402
- Connecting wires
- Bread board.

THEORY: Canonical Forms (Normal Forms): Any Boolean function can be written in disjunctive normal form (sum of min-terms) or conjunctive normal form (product of max-terms). A Boolean function can be represented by a Karnaugh map in which each cell corresponds to a minterm. The cells are arranged in such a way that any two immediately adjacent cells correspond to two minterms of distance 1. There is more than one way to construct a map with this property.

For the above three problems the report should cover: -

- Objective
- Truth table
- Required equation in minimized form with necessary steps
- Circuit diagram
- Required ICs
- IC pin connections in the diagram
- Discussions

Experiment No: 03

Experiment name: Using De Morgan:

a) Design a circuit to verify De Morgan's Law for two variables.

Design a circuit using only a) NAND gates b) NOR gates that implements the Boolean function $(A \oplus B) + (BC)$

AIM: - To Verify Demorgan's theorems.

Apparatus Required:-

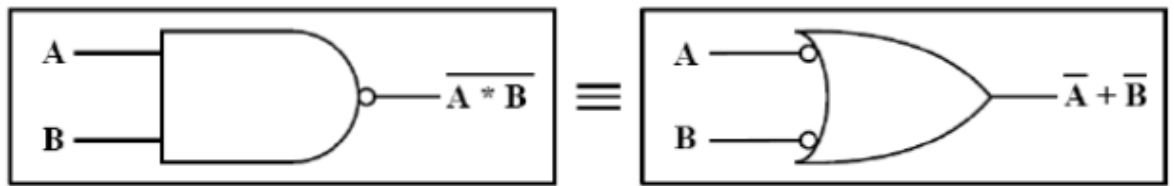
- Breadboard
- Patch cords
- ICs (7404 – NOT, 7408 – AND, 7432 - OR).

Theory:

De Morgan's First Theorem:

- The complement of a product of variables is equal to the sum of the complement of the variables.
- In another way, the complement of two or more ANDed variables is equivalent to the OR of the complements of the individual variables.

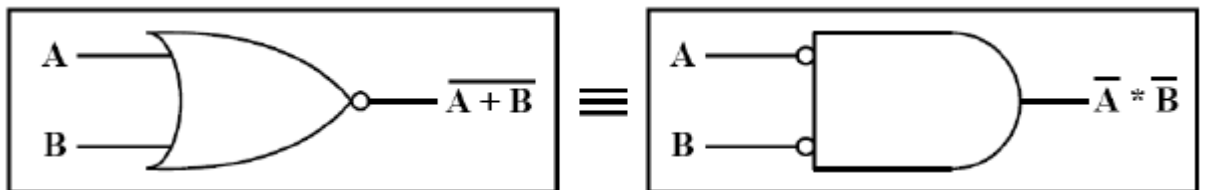
$$(A \cdot B)' = A' + B'$$



De Morgan's Second Theorem:

- The complement of a sum of variables is equal to the product of the complements of the variables.
- In another way, the complement of two or more ORed variables is equivalent to the AND of the complements of the individual variables.

$$(A + B)' = A' \cdot B'$$



Truth Table:

To verify De Morgan's First Theorem: $(A \cdot B)' = A' + B'$

L.H.S. $(A \cdot B)'$			R.H.S. $A' + B'$		
A	B	Y	A	B	Y
0	0	1	0	0	1
0	1	1	0	1	1
1	0	1	1	0	1
1	1	0	1	1	0

To verify De Morgan's Second Theorem: $(A + B)' = A' \cdot B'$

L.H.S. $(A + B)'$			R.H.S. $A' \cdot B'$		
A	B	Y	A	B	Y
0	0	1	0	0	1
0	1	0	0	1	0
1	0	0	1	0	0
1	1	0	1	1	0

Procedure:

Collect the components necessary to accomplish this experiment.

1. Plug the IC chip into the breadboard.
2. Connect the supply voltage and ground lines to the chips. PIN7 = Ground and PIN14 = +5V.
3. According to the pin diagram of each IC mentioned above, make the connections according to circuit diagram.
4. Connect the inputs of the gate to the input switches of the LED.
5. Connect the output of the gate to the output LEDs.
6. Once all connections have been done, turn on the power switch of the breadboard
7. . Operate the switches and fill in the truth table (Write "1" if LED is ON and "0" if LED is OFF Apply the various combination of inputs according to the truth table and observe the condition of Output LEDs.

RESULT: -

Experiment: 04

Experiment Name:

Derive the output equations for a 3 Boolean Function Minimization:

Design and implement a circuit using basic gates that would take a 4-bit binary as input and produce a high level output only if the number is prime or divisible by five.

Design a circuit to implement the following functions (Minimize if necessary)

$$f(A,B,C,D) =$$

$$\sum(2,3,4,6,7,10,11,14)$$

$$f(A,B,C,D) = \Pi(1,3,5,7,9,11)$$

4-bit binary to gray code converter and implement it.

b:

AIM:

To simplify the Boolean expression using K-maps and to build the logic circuit.

Apparatus Required:-

Analog/Digital Training System

- IC Type 7400 Quadruple 2-input NAND gates
- IC Type 7402 Quadruple 2-input NOR gates
- IC Type 7404 Hex Inverters
- IC Type 7408 Quadruple 2-input AND gates
- IC Type 7432 Quadruple 2-input OR gates
- IC Type 7486 Quadruple 2-input XOR gate

Theory:

i. Simplify the equation using K-map and implement it. $f(A,B,C,D) = \sum(2,3,4,6,7,10,11,14)$ and $f(A,B,C,D) = \Pi(1,3,5,7,9,11)$.

ii. Design a digital circuit that can detect a 4-bit prime number.

iii. Design an odd parity generator circuit for 4-bit data.

For the above three problems the report should cover: -

- Objective
- Truth table
- Required equation in minimized form with necessary steps

- Circuit diagram
- Required ICs
- IC pin connections in the diagram
- Discussions

c:

AIM: To realize Binary to Gray code converter.

LEARNING OBJECTIVE:

1. To learn the importance of non-weighted code.
2. To learn to generate gray code.

Apparatus Required:-

- IC 7400
- IC 7486
- IC 7408
- Patch Cords
- IC Trainer Kit.

Truth Table:

Binary				Gray Code			
b ₃	b ₂	b ₁	b ₀	g ₃	g ₂	g ₁	g ₀
0	0	0	0	0	0	0	0
0	0	0	1	0	0	0	1
0	0	1	0	0	0	1	1
0	0	1	1	0	0	1	0
0	1	0	0	0	1	1	0
0	1	0	1	0	1	1	1
0	1	1	0	0	1	0	1
0	1	1	1	0	1	0	0
1	0	0	0	1	1	0	0
1	0	0	1	1	1	0	1
1	0	1	0	1	1	1	1
1	0	1	1	1	1	1	0
1	1	0	0	1	0	1	0
1	1	0	1	1	0	1	1
1	1	1	0	1	0	0	1
1	1	1	1	1	0	0	0

K-map:

For g₀:

		b1,b0			
		00	01	11	10
b3,b2	00	0	1	0	1
	01	0	1	0	1
	11	0	1	0	1
	10	0	1	0	1

For g1:

		b1,b0			
		00	01	11	10
b3,b2	00	0	0	1	1
	01	1	1	0	0
	11	1	1	0	0
	10	0	0	1	1

For g2:

		b1,b0			
		00	01	11	10
b3,b2	00	0	0	0	0
	01	1	1	1	1
	11	0	0	0	0
	10	1	1	1	1

For g3:

		b1,b0			
		00	01	11	10
b3,b2	00	0	0	0	0
	01	0	0	0	0
	11	1	1	1	1
	10	1	1	1	1

Corresponding minimized boolean expressions for gray code bits –

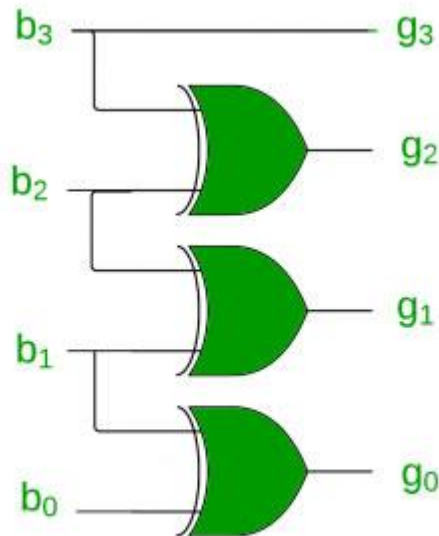
g0= b0(+)b1

g1= b1(+)b2

g2= b2(+)b3

$$g_3 = b_3$$

The corresponding digital circuit –



Procedure:

1. Check all the components for their working.
2. Insert the appropriate IC into the IC base.
3. Make connections as shown in the circuit diagram.
4. Verify the truth table and observe the outputs.

Result:

Experiment: 05

Experiment Name:

Combinational Circuits:

a. Design and implement a circuit using basic gates that would take a 4-bit binary number as input and convert it into Excess - 3

Design and implement a 4-bit adder subtractor circuit with one selection variable to switch for addition and subtraction.

AIM: 1. To realize 4-bit Binary to Excess-3 code converter.

2. To realize 4-bit adder/subtractor circuit with one selection variable to switch for addition and subtraction.

LEARNING OBJECTIVE:

To learn the importance of non-weighted code

To learn to generate Excess-3 code

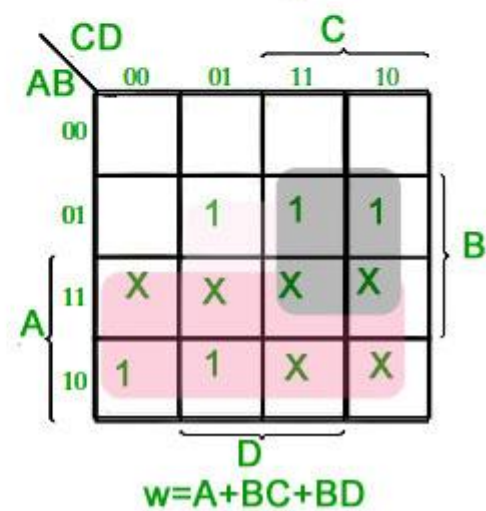
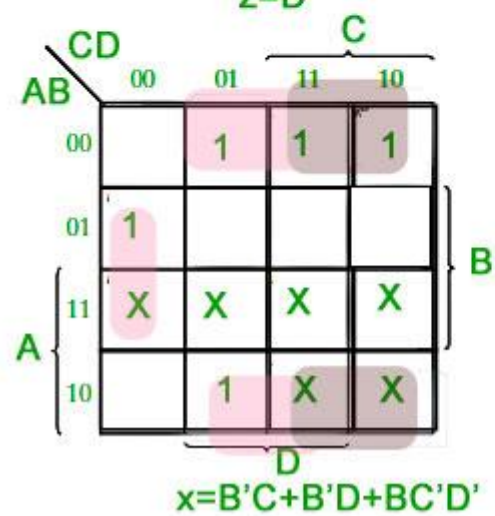
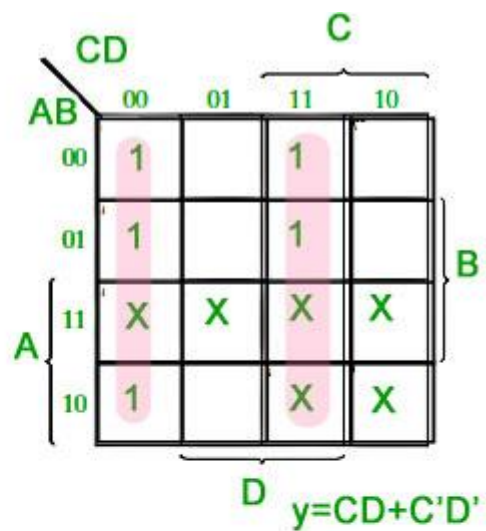
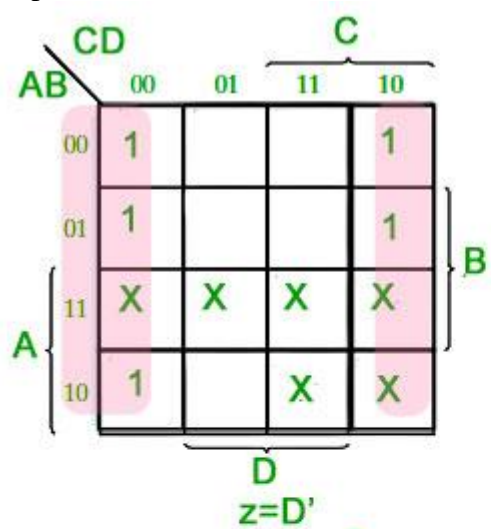
Apparatus Required:-

1. To realize 4-bit Binary to Excess-3 code converter.

Truth Table:

BCD(8421)				Excess-3			
A	B	C	D	w	x	y	z
0	0	0	0	0	0	1	1
0	0	0	1	0	1	0	0
0	0	1	0	0	1	0	1
0	0	1	1	0	1	1	0
0	1	0	0	0	1	1	1
0	1	0	1	1	0	0	0
0	1	1	0	1	0	0	1
0	1	1	1	1	0	1	0
1	0	0	0	1	0	1	1
1	0	0	1	1	1	0	0
1	0	1	0	X	X	X	X
1	0	1	1	X	X	X	X
1	1	0	0	X	X	X	X
1	1	0	1	X	X	X	X
1	1	1	0	X	X	X	X
1	1	1	1	X	X	X	X

K-map:



Circuit Diagram:

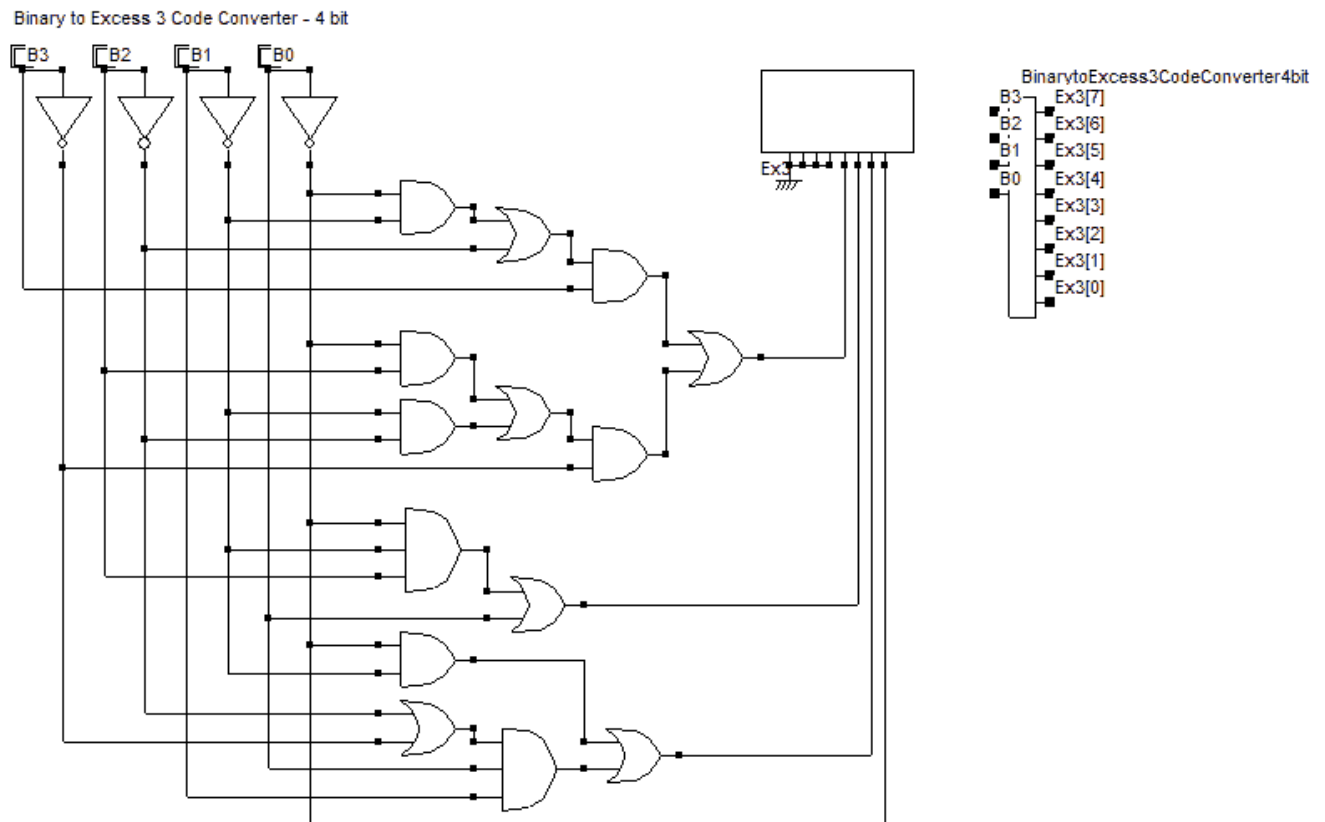


Fig: 4-bit binary to excess-3

2. To realize a 4-bit adder/subtractor circuit with one selection variable to switch for addition and subtraction

4 bit Adder/Subtractor Circuit :

Basic Theory:

It is possible to make a logical circuit that can do both addition and subtraction based on the mode selection concept.

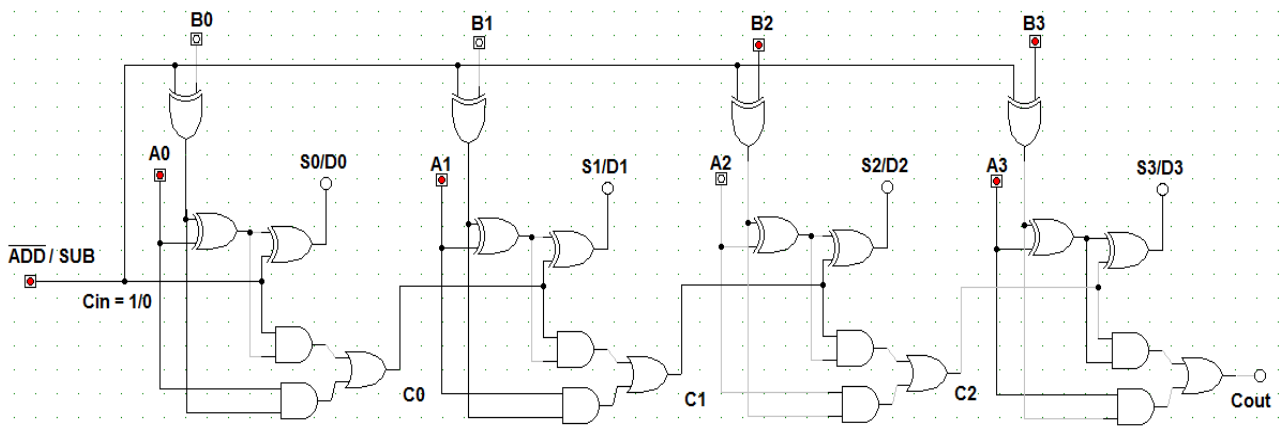
In case of a four bit adder-subtractor, two four bit binary numbers are added or subtracted on the basis of the operation mode. The main principle behind it is subtraction using the two's complement method.

The expression for this is:

$$D_n = A - B = A + B' + 1.$$

This kind of circuit is a ripple-carry adder circuit along with some additional XOR gates to add the subtraction functionality. It works as a full adder if the selected mode is 0 "zero", and works as a full subtractor if the mode is selected as 1 "one". The logic diagram can be seen below.

Circuit Diagram:



Observed Values:

Following set of values were obtained in observation.

Addition:

1. $1.0011 + 0100 = 0111$

2. $1.1101 + 0001 = 1110$

3. $1.1101 + 0010 = 1111$

4. $0.0010 + 0010 = 0100$

5. $1.000 + 1.000 = 0000$; Cout = 1

Subtraction,

1. $1.1011 - 1.100 = 1111$

2. $1.1010 - 0.101 = 0101$

3. $1.1111 - 1.101 = 0010$

4. $0.0011 - 0.011 = 0000$

5. $1.000 - 0.100 = 0100$

Procedure:

1. Check all the components for their working.
2. Insert the appropriate IC into the IC base.
3. Make connections as shown in the circuit diagram.
4. Verify the truth table and observe the outputs.

RESULT: -

Experiment: 07

Experiment Name: Adders and Decoders

a. Design and implement a 4-bit BCD adder circuit using 4-bit binary adder circuit. Use IC-7483 for your design.

b. Design a 2-bit comparator to compare 2-bit numbers A and B

Q. The circuit should provide 3 output lines to indicate $A > B$, $A = B$ and $A < B$.

Aim: - Implementation of 4-bit parallel adder, using 7483 IC.

Apparatus Required:-

1. Digital Trainer Kit.
2. IC 7483
3. Connecting Wires

a. Design and implement a 4-bit BCD adder circuit using 4-bit binary adder circuit. Use IC-7483 for your design.

Theory:-

Adder: - An adder is a logic circuit which adds two or three bits at a time and give sum and carry as the result.

Parallel Adder:-

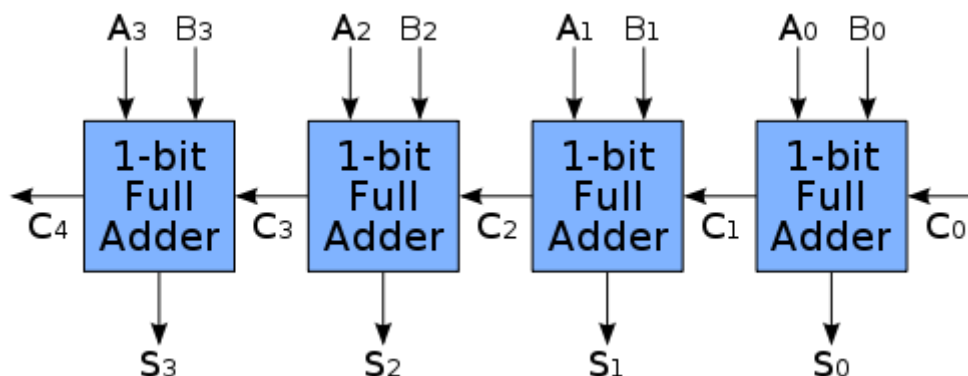
A n-bit parallel adder can be constructed using number of full adders circuit connected in parallel the carry output of each is connected to the carry input of the next higher-order adder. Since all the bits of the augends and addend are fed into the adder circuits simultaneously and the additions in each position are known as parallel adder.

A3 A2 A1 A0 → Augends bits

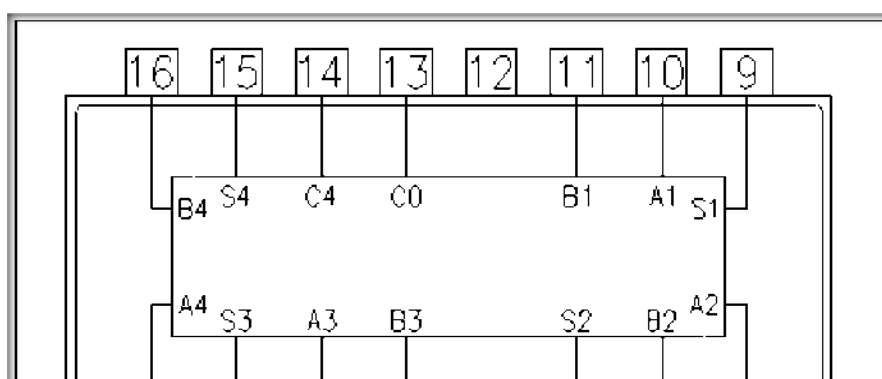
B3 B2 B1 B0 → Addend bits

S3 S2 S1 S0 → Sum bits

LOGIC DIAGRAM OF BCD ADDER



Pin diagram of IC 7483:



Procedure:-

1. Connect ground and Vcc to 7483 IC from trainer kit through patch cords.
2. Connect inputs A0, A1, A2, A3 and B0, B1, B2, B3 to logic input switches.
3. Connect carry in from pin no.13 to ground so that carry input (CY1) will be logic '0'state.
4. Connect S0, S1, S2, S3 and carry out (CY0) from pin nos. 9, 6, 2, 15 and 14 to the output display.
5. Verify truth tables for different combinations of inputs.

TRUTH TABLE:-

The Truth table operation of the 4-bit Parallel Adder is shown below:

INPUTS									OUTPUTS				
A0	A1	A2	A3	B0	B1	B2	B3	CY1	S0	S1	S2	S3	CY0
0	0	0	1	0	1	0	1	1	1	1	0	0	1
1	0	0	0	1	1	1	0	0	0	0	0	0	1

Result:-

For various combinations of selected input lines, observed the LED output and verified the truth table.

b. Design a 2-bit comparator to compare 2-bit numbers A and B

Q. The circuit should provide 3 output lines to indicate $A > B$, $A = B$ and $A < B$.

Theory: A combinational circuit which compares two binary or digital numbers to find out whether the binary number is equal, greater than, less than the other binary number is called Magnitude Comparator Circuit. In this circuit we have 2 inputs A and B. There are 3 outputs available:

1. $A = B$
2. $A > B$
3. $A < B$

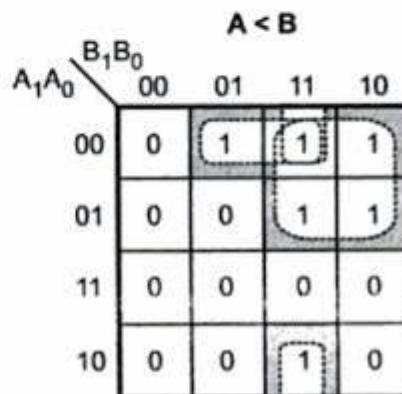
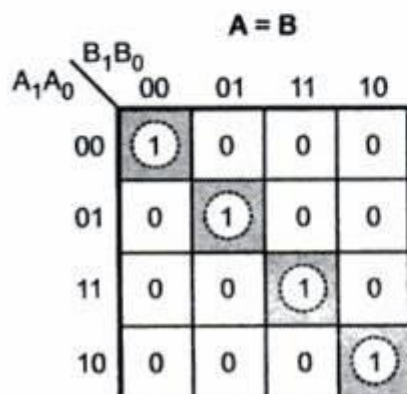
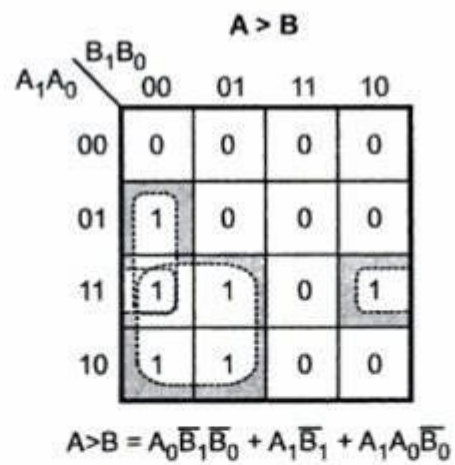
Block diagram of magnitude comparator circuit:



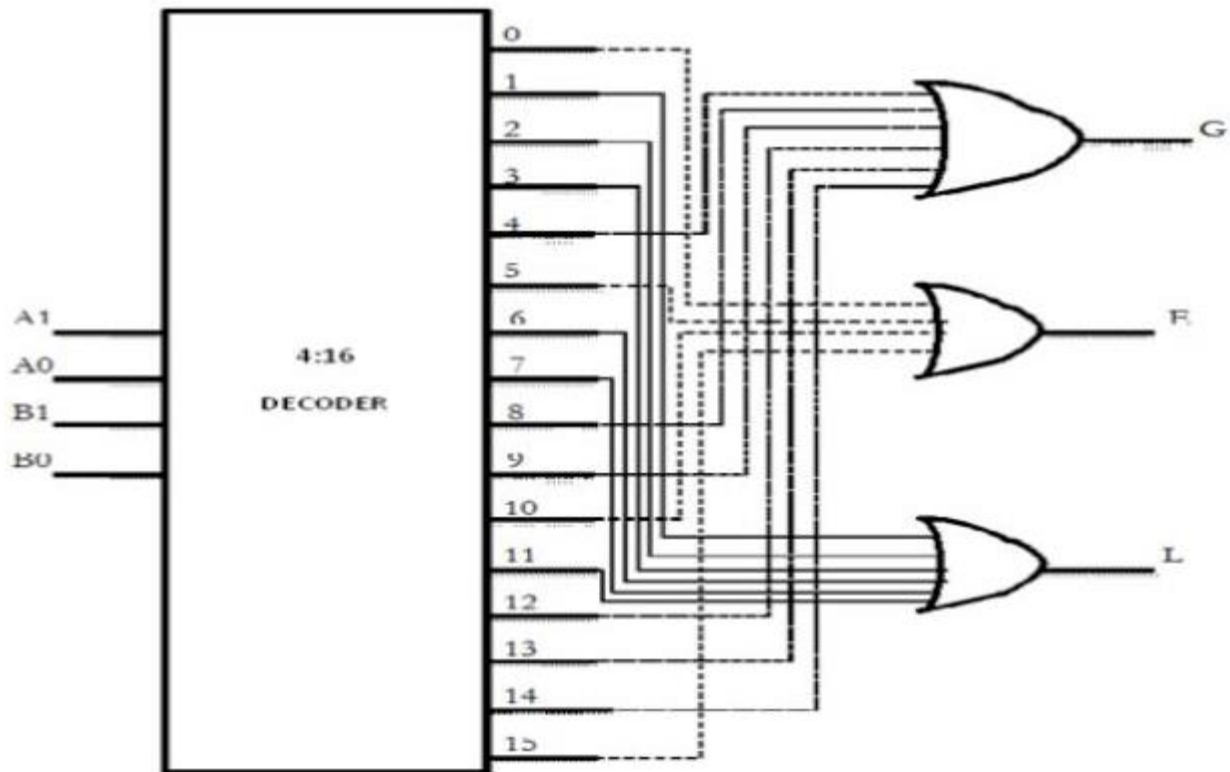
Truth Table:

Inputs				Outputs		
A		B		G(A>B)	E(A=B)	L(A<B)
A1	A0	B1	B0			
0	0	0	0	0	1	0
0	0	0	1	0	0	1
0	0	1	0	0	0	1
0	0	1	1	0	0	1
0	1	0	0	1	0	0
0	1	0	1	0	1	0
0	1	1	0	0	0	1
0	1	1	1	0	0	1
1	0	0	0	1	0	0
1	0	0	1	1	0	0
1	0	1	0	0	1	0
1	0	1	1	0	0	1
1	1	0	0	1	0	0
1	1	0	1	1	0	0
1	1	1	0	1	0	0
1	1	1	1	0	1	0

K-map:



Logic Circuit:



Experiment: 8

Experiment Name:

Priority Encoder and Multiplexer Circuits:

Using basic gates design and implement a 4 ´ 2 priority encoder with the priority of data bit input as given below:

(2, 3, 0, 1 given in order of descending priority)

Design and implement the function given below using 1) 8:1 Multiplexer and 2) 4:1 Multiplexer

$$f = \sum (0, 3, 6, 8, 12)$$

Aim: -

Design and implement simple circuits using multiplexers and encoders.

Apparatus Required:-

- Analog/Digital Training System
- IC Type 74153
- IC Type 74138

Theory:

Solve the following problems:

a. Using basic gates design and implement a 4 ´ 2 priority encoder with the priority of data bit input as given below:

(2, 3, 0, 1 given in order of descending priority)

b. Implement the following functions using multiplexers:

i. $f = \sum (0, 3, 6, 8, 12)$ [Using 4X1 Mux]

ii. $f = \sum (0, 3, 6, 8, 12)$ [Using 8X1 Mux]

For the above three problems the report should cover:

- Objective

- Implementation table (if any)
- Circuit diagram
- Required ICs
- IC pin connections in the diagram
- Discussions

Experiment: 9

Experiment Name: Sequential Circuit Design:

a. Design and implement an S-R flip-flop using only NAND gates. There must be PRESET and CLEAR input.

Design and implement a master-slave J-K flip-flop using only NAND gates.

Aim: - Verification of state tables of

1. R-S flip-flop
 2. Master-slave J - K flip-flop
- Using NAND gate only.

Apparatus Required:-

- IC 7400 (NAND Gate)

Sequential Circuits:

The combinational circuit does not use any memory. Hence the previous state of input does not have any effect on the present state of the circuit. But sequential circuit has memory so output can vary based on input. This type of circuits uses previous input, output, clock and a memory element.

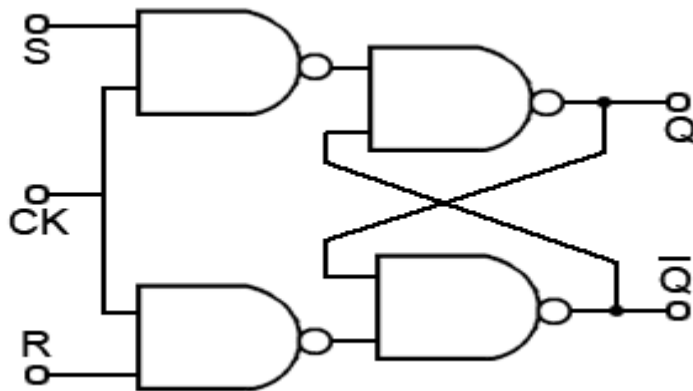
Flip Flop:

Flip flop is a sequential circuit which generally samples its inputs and changes its outputs only at particular instants of time and not continuously. Flip flop is said to be edge sensitive or edge triggered rather than being level triggered like latches.

S-R Flip Flop:

It is basically S-R latch using NAND gates with an additional **enable** input. It is also called as level triggered SR-FF. For this, circuit in output will take place if and only if the enable input (E) is made active. In short this circuit will operate as an S-R latch if $E = 1$ but there is no change in the output if $E = 0$

Logic Diagram



Characteristic table for S-R flip flop

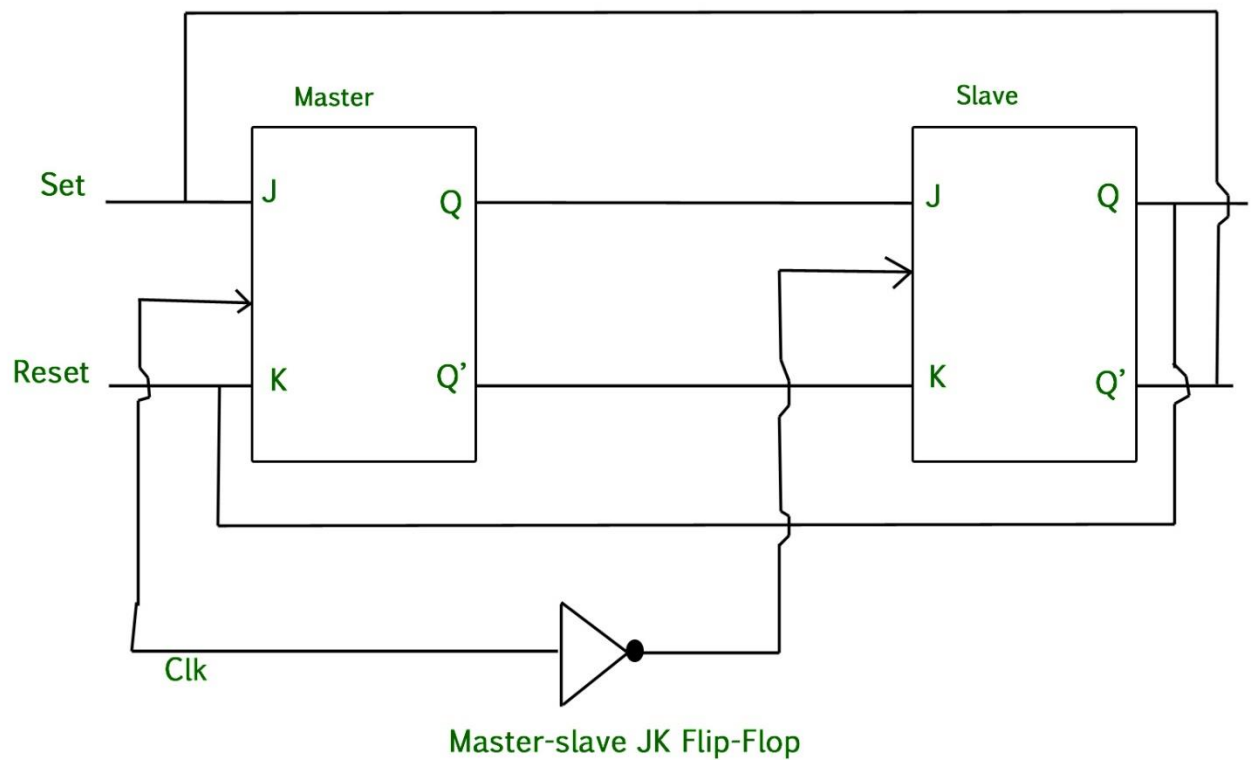
Q	S	R	Q(t+1)
0	0	0	0
0	0	1	0
0	1	0	1
0	1	1	indeterminate
1	0	0	1
1	0	1	0
1	1	0	1
1	1	1	indeterminate

Master Slave JK Flip Flop

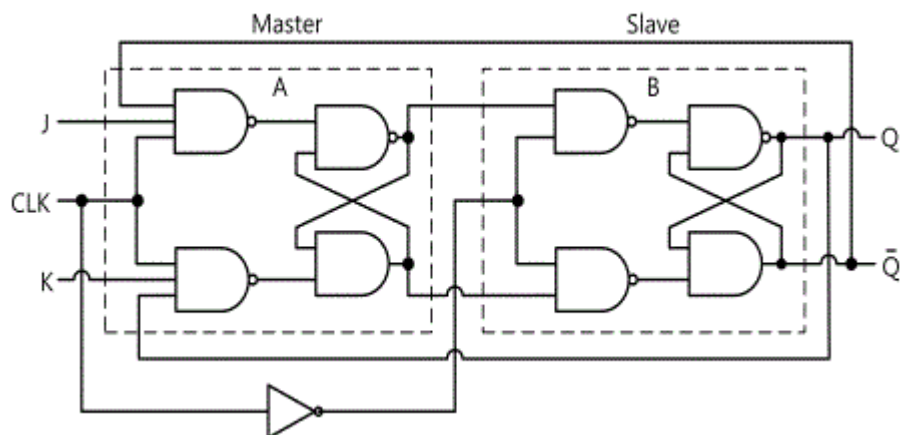
Master slave JK flip-flop could have been designed utilizing 2 JK flip-flops, in that each flip-flop is connected to CLK pulse complementary to each other, and the first flip flop is the master flip-flop which works when the CLK pulse is high state. And at that time the slave flip flop is in the hold state and if the CLK pulse is low state, then the slave flip-flop works, and the master flip-flop stays in the hold state.

The JK flip-flop characteristic is more or less similar to the SR flip-flop, but in SR flip flop, there is one uncertain output state when the S=1 and R =1, but in JK flip flop, when the J=1 and K=1, the flip flop toggles, that means the output state changes from its previous state.

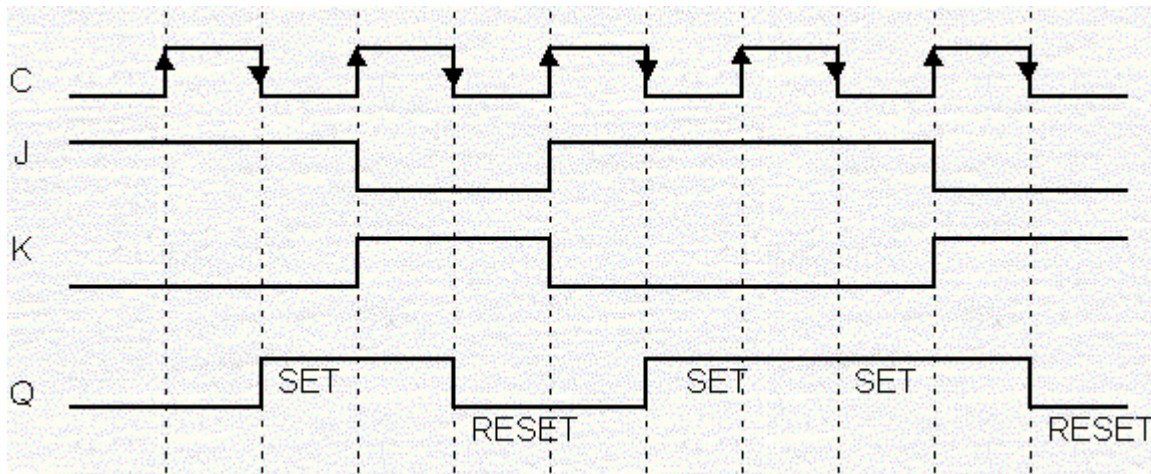
JK Master Slave Flip Flop Circuit Diagram:



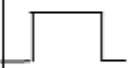
JK Master Slave Flip Flop Circuit Diagram with NAND gate:



JK Master Slave Flip Flop timing Diagram:



JK Master Slave Flip Flop truth table:

		Inputs		Output	
PR	CLR	CLK	J	K	Q
0	0	X	X	X	Race Condition
0	1	X	X	X	1
1	0	X	X	X	0
1	1	X	0	0	No change
1	1		0	1	0
1	1		1	0	1
1	1		1	1	Toggle

Procedure:-

1. Connections are made as per as circuit diagram.
2. Verify truth table for various combinations of input.

RESULT: - Study and verified truth-tables of R-S flip-flop and master-slave JK flip flop.

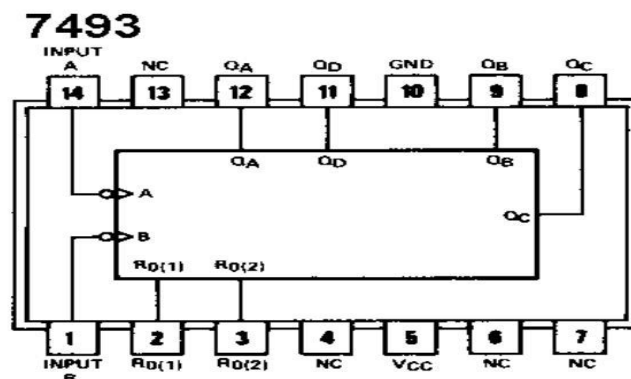
Experiment: 10**Experiment Name:****Counter Circuit Design:****a. Design and implement a 4-bit asynchronous counter using J-K flip flop.****Design and implement a modulo-13 counter using 4-bit binary ripple counter (7493).****AIM:** - To design and verify 4 bit ripple (asynchronous counter).**APPARATUS REQUIRED:-**

S.No.	COMPONENT	SPECIFICATION	QTY.
1.	JK FLIP FLOP	IC 7493	2
2.	NAND GATE	IC 7400	1
3.	IC TRAINER KIT	-	1
4.	PATCH CORDS	-	30

THEORY:-

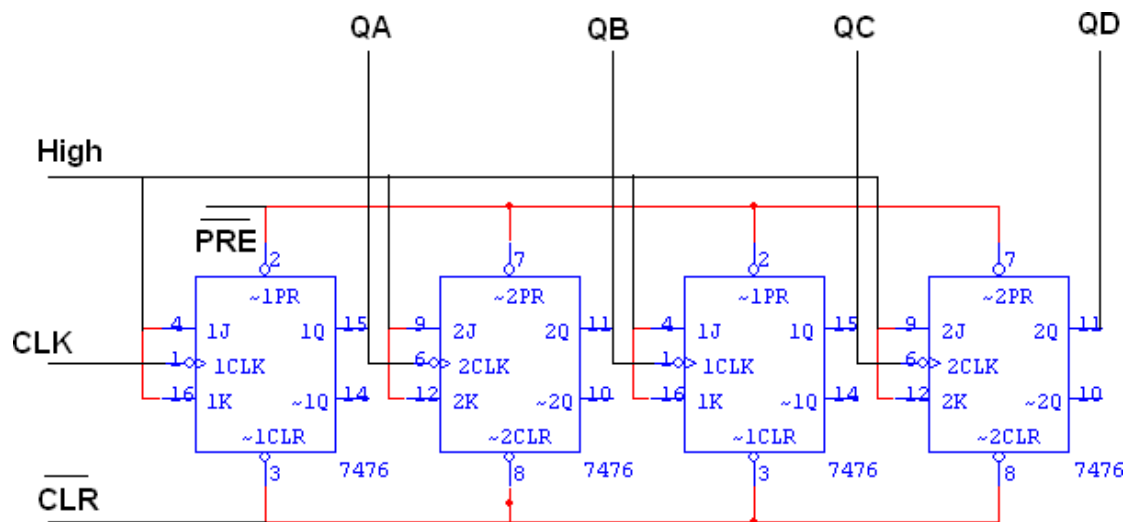
A counter is a register capable of counting number of clock pulse arriving at its clock input. Counter represents the number of clock pulses arrived. A specified sequence of states appears as counter output. This is the main difference between a register and a counter. There are two types of counter, synchronous and asynchronous. In synchronous common clock is given to all flip flop and in asynchronous first flip flop is clocked by external pulse and then each successive flip flop is clocked by Q or Q output of previous stage. As soon the clock of second stage is triggered by output of first stage. Because of inherent propagation delay time all flip flops are not activated at same time which results in asynchronous operation.

IC diagram for 7493



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LOGIC DIAGRAM FOR 4 BIT RIPPLE COUNTER:



TRUTH TABLE:-

CLK	QA	QB	QC	QD
0	0	0	0	0

1	1	0	0	0
2	0	1	0	0
3	1	1	0	0
4	0	0	1	0
5	1	0	1	0
6	0	1	1	0
7	1	1	1	0
8	0	0	0	1
9	1	0	0	1
10	0	1	0	1
11	1	1	0	1
12	0	0	1	1
13	1	0	1	1
14	0	1	1	1
15	1	1	1	1

Procedure:-

- (i) Connections are given as per circuit diagram.
- (ii) Logical inputs are given as per circuit diagram.
- (iii) Observe the output and verify the truth table.

Result: -

Study of 4 bit synchronous counter and verified its truth table

Experiment: 11**Experiment Name: Stopwatch and Counter Circuit Design**

a. Design and implement a digital stopwatch showing only seconds.

Use 7-segment display to show the stopwatch output. You should also provide START, STOP and CLEAR switch.

Design and implement a 4-bit synchronous up-down counter using J-K flip-flops.

Aim: - Design, and verify the 4-bit synchronous up-down counter.

Apparatus Required:-

S.No.	COMPONENT	SPECIFICATION	QTY.
1.	JK FLIP FLOP	IC 7476	2
2.	4 I/P AND GATE	IC 7411	1
3.	OR GATE	IC 7432	1
4.	XOR GATE	IC 7486	1
5.	NOT GATE	IC 7404	1
6.	IC TRAINER KIT	-	1
7.	PATCH CORDS	-	35

Theory:-

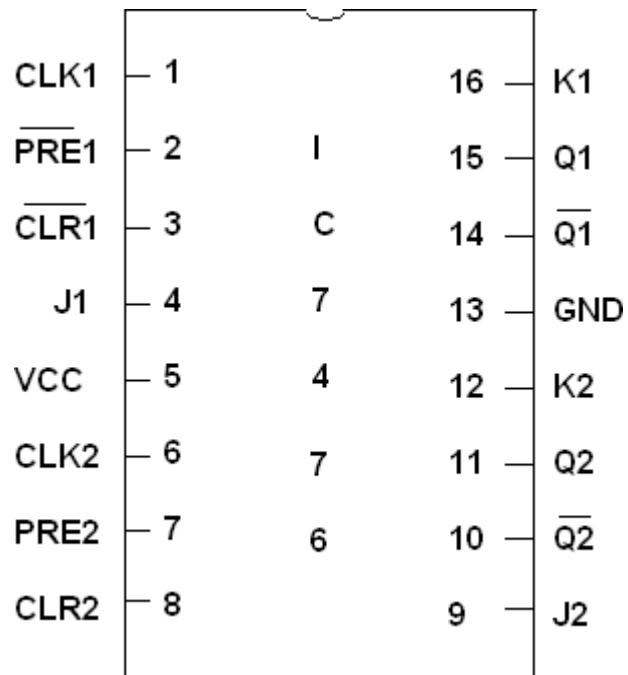
Synchronous counter: - A simple way of implementing the logic for each bit of an ascending counter (which is what is depicted in the image to the right) is for each bit to toggle when all of the less significant bits are at a logic high state. For example, bit 1 toggles when bit 0 is logic high; bit 2 toggles when both bit 1 and bit 0 are logic high; bit 3 toggles when bit 2, bit 1 and bit 0 are all high; and so on.

Synchronous counters can also be implemented with hardware finite state machines.

Which are more complex but allow for smoother, more stable transitions?

Hardware-based counters are of this type and they can be implemented using the IC 7476

PIN DIAGRAM FOR IC 7476:

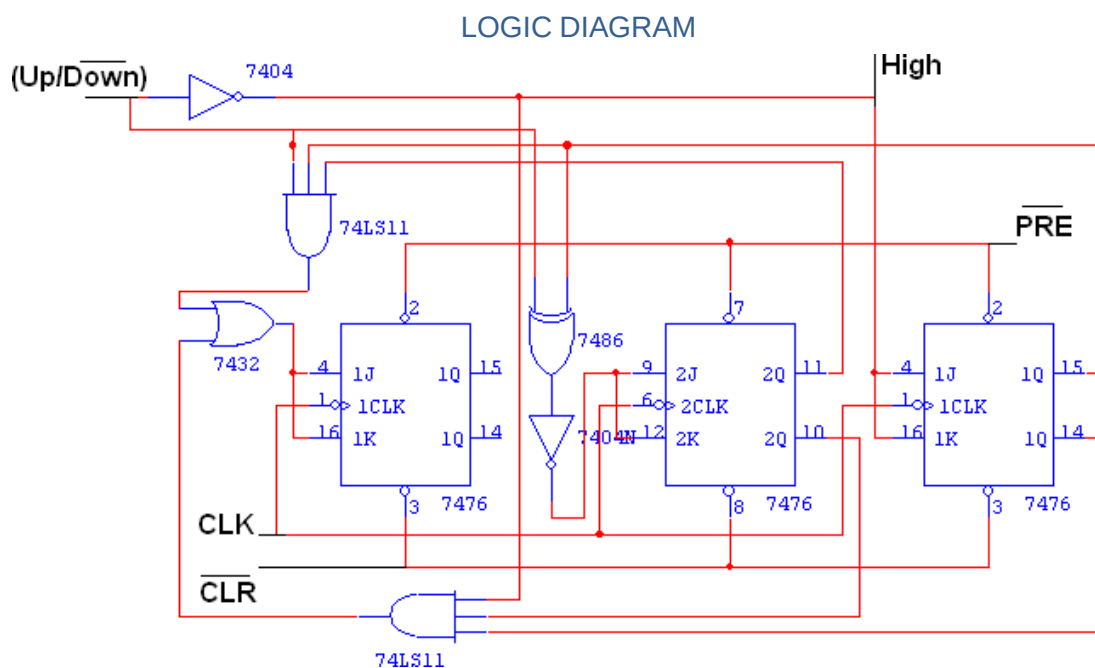


TRUTH TABLE:-

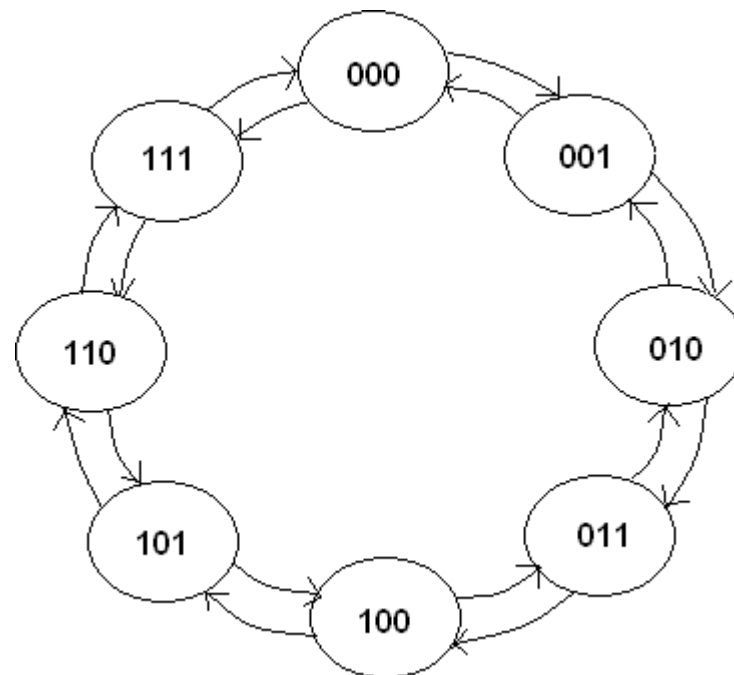
Input	Present State			Next State			A		B		C	
Up/Down	Q _A	Q _B	Q _C	Q _{A+1}	Q _{B+1}	Q _{C+1}	J _A	K _A	J _B	K _B	J _C	K _C
0	0	0	0	1	1	1	1	X	1	X	1	X
0	1	1	1	1	1	0	X	0	X	0	X	1
0	1	1	0	1	0	1	X	0	X	1	1	X
0	1	0	1	1	0	0	X	0	0	X	X	1
0	1	0	0	0	1	1	X	1	1	X	1	X
0	0	1	1	0	1	0	0	X	X	0	X	1
0	0	1	0	0	0	1	0	X	X	1	1	X
0	0	0	1	0	0	0	0	X	0	X	X	1

1	0 0 0	0 0 1	0 X	0 X	1 X
1	0 0 1	0 1 0	0 X	1 X	X 1
1	0 1 0	0 1 1	0 X	X 0	1 X
1	0 1 1	1 0 0	1 X	X 1	X 1
1	1 0 0	1 0 1	X 0	0 X	1 X
1	1 0 1	1 1 0	X 0	1 X	X 1
1	1 1 0	1 1 1	X 0	X 0	1 X
1	1 1 1	0 0 0	X 1	X 1	X 1

TRUTH TABLE OF 4 BIT SYNCHRONOUS COUNTERS



State diagram



Procedure:-

- (i) Connections are given as per circuit diagram.
- (ii) Logical inputs are given as per circuit diagram.
- (iii) Observe the output and verify the truth table.

Result: -Study of 4 bit synchronous counters and verified its truth table.